

Mohammadmahdi Maharebi

M.Sc. Student / Electrical Communication Engineering

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Profile

M.Sc. student in Electrical Communication Engineering focused on software development, C++/Python scientific computing, simulation tools, and technical documentation. Builds public portfolio projects that combine numerical methods, real-time visualization, reproducible workflows, and clear GitHub documentation. Background in electromagnetics, RF engineering, and signal processing; silver medalist in Iran's National Physics Olympiad. Interested in software working-student roles, student assistant / HiWi positions, scientific computing, and engineering software development.

Education

- 2024–Present **M.Sc. Electrical Communication Engineering**, University of Kassel, Kassel, Germany, GPA: 1.53/1.0 (*sehr gut*)
Key topics: microwaves, optoelectronics, engineering mathematics, digital communications and lab. Focus on computational EM, photonics, and optimization methods.
- 2017–2023 **B.Sc. Electrical Engineering**, Sharif University of Technology, Tehran, Iran
Emphasis on electromagnetics, microwave engineering, signal processing, and programming coursework.

Selected Software & Engineering Projects

- 2026–Present **Symulation – Interactive C++ Simulation Lab**, C++17 / CMake / raylib / ImGui / ImPlot / ImPlot3D / Eigen / CGAL, Public portfolio project
- Built a native desktop simulation lab for physics, electromagnetics, and numerical visualization with real-time GUI controls and a modular architecture.
 - Implemented modules for BCC spring-mass wave propagation, Hertzian dipole fields, thin-wire dipole MoM/Pocklington demo, double pendulum, three-body gravity, and Helmholtz FEM eigenmodes on editable Bézier domains.
 - Prepared public release with Git submodules, build instructions, GIF demos, technical notes, MIT license, GitHub documentation, portfolio integration, and a technical blog post.
 - Code: github.com/mmaharebi/symulation; Blog: mahdymahareb.de/blog/symulation
- 2024–Present **Personal Portfolio Website**, Next.js / React / TypeScript / MDX, Public website
- Built and maintain a portfolio site for projects, technical writing, and public documentation.
 - Uses typed React components and MDX content to present engineering projects, notes, demos, and links.
 - Website: mahdymahareb.de
- 2023 **Adaptive FEM Visualization Tool**, C# / WPF, Engineering software
- Implemented a desktop tool for FEM field visualization and mesh reconstruction.
 - Integrated Delaunay triangulation and automated parsing routines for simulation data.
- 2025 **2D FDTD PEC Cavity Solver**, Python / NumPy, Computational electromagnetics
- Implemented a TM_z FDTD solver on a Yee lattice and validated five cavity modes with mean error **0.43%**.
 - Added FFT-based eigenfrequency extraction, convergence plots, animations, and reproducible scripts.
 - Code: github.com/mmaharebi/fdtd-pec-cavity

Technical Skills

Languages	C++17, Python, TypeScript/React/Next.js, C#, Java, MATLAB
Software Tools	Git/GitHub, Linux/Bash, CMake, LaTeX, VS Code, Jupyter

Scientific Computing	NumPy, SciPy, Matplotlib, Eigen, CGAL, FDTD, FEM post-processing, eigenmode solvers
Web & Documentation	React components, MDX content, technical notes, public GitHub repositories
EM / RF	HFSS, ADS; S-parameters, Smith chart, array factor analysis, antenna and microwave circuit basics

Teaching & Experience

2021–2023	Teaching Assistant , <i>Sharif University of Technology</i> , Tehran, Iran Circuit Analysis I (<i>Head TA</i>), Circuit Analysis II, and Object-Oriented Programming in Java. Supported labs, grading, problem solving, and student mentoring across electronics and programming coursework.
2019	Intern / Project Contributor – 5G Demo Platform , <i>Hamrah-e Avval (MCI Iran) / Sharif University of Technology</i> , Tehran, Iran Built a gesture-controlled 5G demonstration using Arduino and Leap Motion sensor as part of a university-supervised project for Hamrah-e Avval (MCI Iran). Presented low-latency communication concepts in a robotics setup for visitors and internal stakeholders.

Awards

2017	Silver Medal — Iran's National Physics Olympiad
2017	National Elites Foundation Recognition (Iran)

Languages

English	Professional proficiency	<i>Comfortable with technical communication and documentation</i>
German	A2.1 (actively improving)	<i>Learning; familiar with technical coursework</i>
Persian	Native	<i>Mother tongue</i>

Links

GitHub	github.com/mmaharebi
Portfolio	mahdymahareb.de
Key Repos	Symulation, portfolio website, FDTD cavity solver, FEM visualization tool